

**PART-I**

S.No: B304091



Programme : B.Tech

Hall Ticket No: 24B11AI240

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REGULAR

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Signature of the Controller of Examination

Signature of the Student with date

Signature of the Invigilator with date

Serial No.
Last Page Written

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INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
 - i. Writing their Hall Ticket No. and name in any part of the answer booklet.
 - ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
 - iii. Writing religious symbols.
 - iv. Either seeking or providing any assistance to the fellow students in the examinations.
 - v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
 - vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
 - vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**



SPACE FOR ROUGH WORK

25/1/21

25/1/21

25



2a)

Given,

$$f(x) = kx^2 e^{-x} \text{ for } x \geq 0 \rightarrow 0.$$

we know that,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\rightarrow \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx = 1$$

$$\rightarrow 0 + \int_0^{\infty} kx^2 e^{-x} dx = 1$$

$$\rightarrow k \int_0^{\infty} x^2 e^{-x} dx = 1$$

$$\rightarrow k \left[x^2 \left[\frac{e^{-x}}{-1} \right] - 2x \left[\frac{e^{-x}}{(-1)^2} \right] + 2 \left[\frac{e^{-x}}{(-1)^3} \right] \right]_0^{\infty} = 1$$

$$\rightarrow k \left[-x^2 e^{-x} + 2x e^{-x} - 2e^{-x} \right]_0^{\infty} = 1$$

$$\rightarrow k \left[(-0 + 0 - 0) - (-0 + 0 - 2) \right] = 1$$

$$\rightarrow k [2] = 1 \rightarrow k = \frac{1}{2}$$

$$\therefore f(x) = \frac{1}{2} x^2 e^{-x} \text{ for } x \geq 0.$$

$$\text{Mean, } \mu = \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^0 0 dx + \int_0^{\infty} x \cdot \frac{1}{2} x^2 e^{-x} dx$$

$$\rightarrow \frac{1}{2} \int_0^{\infty} x^3 e^{-x} dx.$$



$$\int_0^{\infty} \frac{1}{2} x^3 e^{-x} dx = \frac{1}{2} \left[x^3 \left[\frac{e^{-x}}{-1} \right] - 3x^2 \left(\frac{e^{-x}}{-1} \right) + 6x \left(\frac{e^{-x}}{(-1)^3} \right) - 6 \left(\frac{e^{-x}}{(-1)^4} \right) \right]_0^{\infty}$$

$$\Rightarrow \frac{1}{2} [(0 - 0 + 0 - 0) - (0 - 0 + 0 - 6)] = (3)$$

$$\Rightarrow \frac{1}{2}(6) = 3 = \text{Mean}$$

$$\text{Variance, } \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$$

$$\Rightarrow \int_{-\infty}^0 0 dx + \int_0^{\infty} x^2 \cdot \frac{1}{2} x^2 e^{-x} dx$$

$$\Rightarrow \frac{1}{2} \int_0^{\infty} x^4 e^{-x} dx$$

$$\Rightarrow \frac{1}{2} \left[x^4 \left[\frac{e^{-x}}{-1} \right] - 4x^3 \left[\frac{e^{-x}}{(-1)^2} \right] + 12x^2 \left[\frac{e^{-x}}{(-1)^3} \right] - 24x \left[\frac{e^{-x}}{(-1)^4} \right] + 24 \left[\frac{e^{-x}}{(-1)^5} \right] \right]_0^{\infty}$$

$$\Rightarrow \frac{1}{2} [(0 - 0 + 0 - 0 + 0) - (0 - 0 + 0 - 0 - 24)]$$

$$\Rightarrow \frac{1}{2}(24) = 12 = \text{Variance}$$

$$\therefore \text{Mean, } \mu = 3 ; \text{ Variance } \sigma^2 = 12.$$



2b.) Given,

$$F(x) = \begin{cases} 0, & \text{if } x \leq 1 \\ k(x-1)^3, & \text{if } 1 \leq x \leq 3 \\ 1, & \text{if } x > 3. \end{cases}$$

i.) K

We know that, $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_{-\infty}^1 0 dx + \int_1^3 k(x-1)^3 dx + \int_3^{\infty} 1 dx = 1$$

$$\Rightarrow 0 + k \int_1^3 (x-1)^3 dx + \int_3^{\infty} 1 dx = 1$$

$$\Rightarrow k \left[\int_1^3 (x^3 - 1 + 3x - 3x^2) dx \right] + \left[x \right]_3^{\infty} = 1$$

$$\Rightarrow k \left[\frac{x^4}{4} - x + \frac{3x^2}{2} - x^3 \right]_1^3 + \dots = 1$$

$$\Rightarrow k \left[\left(\frac{3^4}{4} - 3 + \frac{3(3)^2}{2} - (3)^3 \right) - \left(\frac{1}{4} - 1 + \frac{3}{2} - 1 \right) \right] + \dots = 1$$

$$\Rightarrow k \left[(20.25 - 3 + 13.5 - 27) - \left(\frac{1}{4} - 2 \right) \right] + \dots = 1$$

$$\Rightarrow k [3.75 + 1.75] = 1$$

$$k [4] = 1 \Rightarrow k = \frac{1}{4}$$

ii.) By substituting K, we get the Pdf.

$$\therefore \text{P.d.f of } X = \begin{cases} 0 & \text{if } x \leq 1 \\ \frac{1}{4}(x-1)^3 & \text{if } 1 \leq x \leq 3 \\ 1 & \text{if } x > 3. \end{cases}$$



$$\text{iii.) Mean, } \mu = \int_{-\infty}^{\infty} x f(x) dx$$

$$\Rightarrow 0 + \frac{1}{4} \int_{-2}^3 x(x-1)^3 dx + 1$$

$$\Rightarrow 0 + \frac{1}{4} \int_{-2}^3 x [x^3 - 1 + 3x - 3x^2] dx$$

$$\Rightarrow \frac{1}{4} \int_{-2}^3 [x^4 - x + 3x^2 - 3x^3] dx$$

$$\Rightarrow \frac{1}{4} \left[\frac{x^5}{5} - \frac{x^2}{2} + \frac{3x^3}{3} - \frac{3x^4}{4} \right]_{-2}^3$$

$$\Rightarrow \frac{1}{4} \left[\frac{x^5}{5} - \frac{x^2}{2} + \frac{3x^3}{3} - \frac{3x^4}{4} \right]_1^3$$

$$\Rightarrow \frac{1}{4} \left[\frac{x^5}{5} - \frac{x^2}{2} + x^3 - \frac{3x^4}{4} \right]_1^3$$

$$\Rightarrow \frac{1}{4} \left[\left(\frac{3^5}{5} - \frac{3^2}{2} + 3^3 - \frac{3(3)^4}{4} \right) - \left(\frac{1}{5} - \frac{1}{2} + 1 - \frac{3}{4} \right) \right]$$

$$\Rightarrow \frac{1}{4} \left[\left(\frac{243}{5} - \frac{9}{2} + 27 - \frac{243}{4} \right) - (0.2 - 0.5 + 1 - 0.75) \right]$$

$$\Rightarrow \frac{1}{4} \left[(48.6 - 4.5 + 27 - 60.75) - (-0.05) \right]$$

$$\Rightarrow \frac{1}{4} [10.35 + 0.05] \Rightarrow \frac{10.4}{4} = 2.6$$

$$\therefore \text{Mean, } \mu = 2.6$$



3a.) Given,

$$\text{mean, } \mu = 78$$

$$\text{Standard deviation, } \sigma = 11.$$

$$\text{number of students } n = 1000.$$

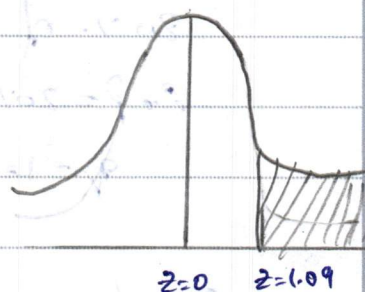
$$\therefore Z = \frac{x - \mu}{\sigma} = \frac{x - 78}{11}$$

i) no. of students got marks above 90.

$$x = 90.$$

$$Z = \frac{x - \mu}{\sigma} = \frac{90 - 78}{11} = \frac{12}{11} = 1.09$$

$$\begin{aligned} P(x > 90) &= P(Z > 1.09) \\ &= 0.5 - A(1.09) \\ &= 0.5 - 0.3621 \\ &= 0.1379. \end{aligned}$$



For 1000 students,

$$\text{no. of students got marks above 90} = 0.1379 \times 1000$$

$$= 137.9$$

$$\approx 138 \text{ students.}$$

$\therefore$ no. of students who got marks above 90 = 138 students.

ii) no. of students who got marks below 60.

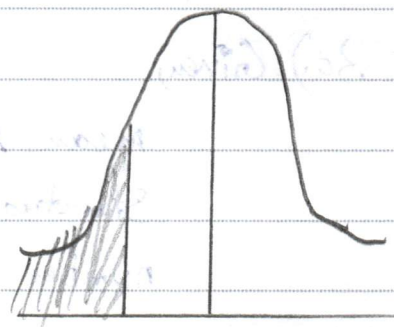
$$x = 60,$$

$$Z = \frac{x - \mu}{\sigma} = \frac{60 - 78}{11} = \frac{-18}{11} = -1.636 \approx -1.64.$$

$$P(x < 60) = P(Z < -1.64)$$



$$\begin{aligned}
 P(z < -1.636) &= 0.5 - A(1.636) \\
 &= 0.5 - 0.4495 \\
 &= 0.0505.
 \end{aligned}$$



For 1000 students,
 no. of students who got below 60 marks
 $= 0.0505 \times 1000$
 $= 50.5$
 ≈ 50 students.

3b.) Given,

20% of items produced from a factory are defective.

$$p = 20\% \text{ (or) } 0.2 \text{ (or) } \frac{1}{5}$$

$$q = 1 - p = 1 - 0.2 = 0.8 \text{ (or) } \frac{4}{5}$$

Size of sample $n = 5$.

∴ By binomial distribution,

$$P(X = r) = {}^5C_r \left(\frac{1}{5}\right)^r \left(\frac{4}{5}\right)^{5-r}.$$

i) none is defective.

$$P(X = 0) = {}^5C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{5-0}$$

$$= {}^5C_0 (1) \left(\frac{4}{5}\right)^5 = \frac{1024}{3125}$$

$$\therefore P(X = 0) = 1(1)(0.32768) = 0.32768$$



∴ Probability of none is defective $P(X=0) = 0.32768$.

ii) One is defective.

$$\begin{aligned} P(X=1) &= {}^5C_1 \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^{5-1} \\ &= 5 \left(\frac{1}{5}\right) \left(\frac{256}{625}\right) \end{aligned}$$

$$\therefore P(X=1) = 0.4096.$$

Probability of only one defective item $P(X=1) = 0.4096$.

iii) at most 2 defective,

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= {}^5C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^5 + {}^5C_1 \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^4 + {}^5C_2 \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^3$$

$$= 1(1) \left(\frac{4}{5}\right)^5 + 5 \left(\frac{1}{5}\right) \left(\frac{4}{5}\right)^4 + 10 \left(\frac{1}{25}\right) \left(\frac{64}{125}\right)$$

$$= 0.32768 + 0.4096 + 0.2048.$$

$$= 0.94208.$$

∴ Probability of at most 2 items being defective is,

$$P(X \leq 2) = 0.94208.$$



5a)

In data there are two types of analysis
They are:-

- i.) Qualitative analysis
- ii.) Quantitative analysis

i.) Qualitative analysis:-

This is based on the quality or characteristics of the data.

It is of two types

i.) Nominal:-

The data has no order.

ii.) Ordinal:-

The data is using some sort of order.

Eg:- low, high, ...etc.

ii.) Quantitative analysis:-

This is based on the quantity and can be measured.

It is also of two types. They are:-

i.) Discrete:-

They can be counted and exist as separate values. Eg:- 1, 2, 3, ...etc.

ii.) Continuous:-

They cannot be counted and exist similar



to ranges. The no. of values containing cannot be counted and could also be infinity.

Eg:- 0 to 1, 10 to 20... etc.

These are the types of data.

* Methods of data Collection:-

There are two types of data collection. When collecting data, we use any of these two approaches.

They are:-

i) Primary data Collection:-

In this approach, the person collects original data without the data being taken or referred from another source. All of the collection will be done by the person.

ii) Secondary data Collection:-

In this approach, we use some already existing collections to take our data. So, in this approach, the data is collected or derived from already existing data collections.

These are the types of data collections.



56)

Given,

numbers: 2, 3, 6, 8, 11.

no. of elements, $N = 5$ Size of sample, $n = 2$

i.) Mean of the Population,

$$\mu = \frac{\sum x_i}{N} = \frac{2+3+6+8+11}{5} = 6.$$

$$\therefore \mu = 6.$$

ii.) Standard deviation of Population

$$\text{Variance, } \sigma^2 = \frac{\sum (x - \mu)^2}{N} = \frac{(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2}{5}$$

$$= \frac{(-4)^2 + (-3)^2 + (0)^2 + (2)^2 + (5)^2}{5}$$

$$= \frac{16 + 9 + 0 + 4 + 25}{5} = \frac{54}{5} = 10.8$$

$$\therefore \sigma = 10.8.$$



iii) Mean of Sampling distribution of Means.

no. of total samples with replacement $= N^n = 5^2 = 25$.

The samples are,

$(2,2)$ $(2,3)$ $(2,6)$ $(2,8)$ $(2,11)$
 $(3,2)$ $(3,3)$ $(3,6)$ $(3,8)$ $(3,11)$
 $(6,2)$ $(6,3)$ $(6,6)$ $(6,8)$ $(6,11)$
 $(8,2)$ $(8,3)$ $(8,6)$ $(8,8)$ $(8,11)$
 $(11,2)$ $(11,3)$ $(11,6)$ $(11,8)$ $(11,11)$

Sampling distribution means are,

2 2.5 4 5 6.5
 2.5 3 4.5 5.5 7
 4 4.5 6 7 8.5
 5 5.5 7 8 9.5
 6.5 7 8.5 9.5 11

Mean of sampling distribution of Means, $\mu_{\bar{x}}$

$$\begin{aligned}
 \mu_{\bar{x}} = & 2 + 2.5 + 4 + 5 + 6.5 + 2.5 + 3 + 4.5 + 5.5 + 7 + 4 \\
 & + 4.5 + 6 + 7 + 8.5 + 5 + 5.5 + 7 + 8 + 9.5 + \\
 & 6.5 + 7 + 8.5 + 9.5 + 11 \\
 & \hline
 & 25
 \end{aligned}$$



$$= \frac{150}{25} = 6.$$

iv.) Standard deviation of sampling distribution of means,

$$\sigma_{\bar{x}}^2 = \frac{\sum (x - \mu)^2}{N-1}$$

$$= \frac{(2-6)^2 + (2.5-6)^2 + (4-6)^2 + (5-6)^2 + (6.5-6)^2 + (2.5-6)^2 + (3-6)^2 + (4.5-6)^2 + (5.5-6)^2 + (7-6)^2 + (4-6)^2 + (4.5-6)^2 + (6-6)^2 + (7-6)^2 + (8.5-6)^2 + (5-6)^2 + (5.5-6)^2 + (7-6)^2 + (8-6)^2 + (9.5-6)^2 + (6.5-6)^2 + (7-6)^2 + (8.5-6)^2 + (9.5-6)^2 + (11-6)^2}{24}$$

$$= \frac{16 + 12.25 + 4 + 1 + 0.25 + 12.25 + 9 + 2.25 + 0.25 + 1 + 4 + 2.25 + 0 + 1 + 6.25 + 1 + 0.25 + 1 + 4 + 12.25 + 0.25 + 1 + 6.25 + 12.25 + 25}{24}$$

$$= \frac{135}{24} = 5.625.$$

Standard deviation of sampling distribution means,

$$\sigma_{\bar{x}} = \sqrt{\sigma_{\bar{x}}^2} = \sqrt{5.625} = 2.371.$$

∴ Standard deviation of sampling distribution of means

$$\sigma_{\bar{x}} = 2.371$$



7a.)

Given,

size of sample, $n = 49$. $\bar{x} = 15200$ km mean lifemean, $\mu = 15150$ km.Standard deviation, $\sigma = 120$ km.Null hypothesis: $H_0: \mu = 15150$ Alternative hypothesis: $H_1: \mu \neq 15150$ level of significance: $\alpha = 0.05$ (or) 95% ConfidenceCritical value: $z_\alpha = 1.96$ at $\alpha = 0.05$ (or) 95% Confidence

$$\text{Test statistic} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{15200 - 15150}{\frac{120}{\sqrt{49}}}$$

$$= \frac{50}{\frac{120}{7}} = \frac{5 \times 7}{12} = 2.91667$$

Conclusion: $|z| > z_\alpha (=1.96)$

$$\Rightarrow |z| = 2.91667 > z_\alpha (=1.96)$$

Null hypothesis is rejected.

Alternative hypothesis is accepted.

The sample is not drawn from given population.



7b)

Given,

no. of Englishmen, $n_1 = 6400$.mean of Englishmen height, $\bar{x}_1 = 67.85$.Standard deviation of Englishmen height, $\sigma_1 = 2.56$.no. of Austrians, $n_2 = 1600$.mean of Austrians height, $\bar{x}_2 = 68.55$.Standard deviation of Austrian height, $\sigma_2 = 2.52$.Level of significance, $\alpha = 0.01$.Null hypothesis, $H_0: \mu_1 = \mu_2$. Their average height is similar.Alternative hypothesis, $H_1: \mu_1 < \mu_2$. Austrians on average taller than englishmen.level of significance: $\alpha = 0.01$ @ 99% Confidence.Critical region: at $\alpha = 0.01$ @ 99% Confidence,
 $z_\alpha = 1.65$.

$$\text{Test statistic: } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \rightarrow z = \frac{67.85 - 68.55}{\sqrt{\frac{(2.56)^2}{6400} + \frac{(2.52)^2}{1600}}}$$



$$\Rightarrow Z = \frac{-0.7}{\sqrt{\frac{6.5536}{6400} + \frac{6.3504}{1600}}}$$

$$\Rightarrow Z = \frac{-0.7}{\sqrt{1.024 \times 10^{-3} + 3.969 \times 10^{-3}}}$$

$$Z = \frac{-0.7}{\sqrt{4.993 \times 10^{-3}}}$$

$$Z = \frac{-0.7}{0.07066} = -9.906$$

Conclusion:-

$$|Z| = 9.906 > Z_{\alpha} (= 1.65)$$

Null hypothesis is rejected.

Alternative hypothesis is accepted.

$\therefore$ The Austrians are on average taller than the Englishmen.



9a)

Given,

Let the required equation be $y = ae^{bx}$ - (1)

by applying log on both sides,

$$\log y = \log(ae^{bx})$$

$$\rightarrow \log y = \log a + \log e^{bx}$$

$$\rightarrow \log y = \log a + bx \text{ - (2)}$$

Let (2) $\Rightarrow Y = A + BX$, where $y = \log y$, $A = \log a$, $B = b$, $X = x$. - (3)

The required equations are,

$$\Sigma Y = nA + B \Sigma X \text{ - (4)}$$

$$\Sigma XY = \Sigma XA + B \Sigma X^2 \text{ - (5)}$$

Given,

x	y	X=x	Y=log y	X ²	XY
0.0	0.10	0.0	-1	0	0
0.5	0.45	0.5	-0.346	0.25	-0.173
1.0	0.15	1.0	0.332	1	0.332
1.5	0.15	1.5	0.961	2.25	1.441
2.0	40.35	2.0	1.605	4	3.21
2.5	180.75	2.5	2.254	6.25	5.64

$$\Sigma X = 7.5 \quad \Sigma Y = 3.809 \quad \Sigma X^2 = 13.75 \quad \Sigma XY = 10.45$$

By substituting these values in (4) & (5), we get

$$3.809 = 6A + B(7.5) \text{ - (6)}$$

$$10.45 = 7.5A + 13.75B \text{ - (7)}$$



Null Hypothesis : $H_0 : \mu_1 = \mu_2$

Alternative Hypothesis : $H_1 : \mu_1 \neq \mu_2$

level of significance $\alpha = 0.01$ (1%)

critical value $z_{\alpha/2} = z_{0.005}$

$z_{tab} = 2.58$

Test statistic :

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$= \frac{1100 - 900}{\sqrt{\frac{240^2}{100} + \frac{220^2}{200}}}$$

$$= \frac{200}{\sqrt{576 + 242}}$$

$$z = \frac{200}{\sqrt{818}}$$

$$z = \frac{200}{28.6006} = 6.99$$

$$z_{cal} = 6.99$$

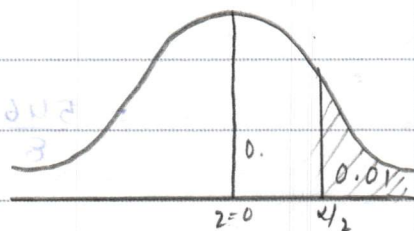
$$z_{cal} > z_{tab}$$

H_0 is not accepted

$\therefore$ z calculated value is greater than

z tabulated value. efficiency in factories is not same

$\therefore H_1$ is accepted.



10a)

$$\text{Karl Pearson's} = \frac{\sum XY}{\sqrt{\sum X^2 \sum Y^2}}$$

$$x = x - \bar{x} \quad , \quad y = y - \bar{y}$$

$$\bar{y} = \frac{\sum y}{n} \quad \bar{x} = \frac{\sum x}{n}$$

$$\bar{x} = \frac{65 + 66 + 67 + 67 + 68 + 69 + 71 + 73}{8}$$

$$= \frac{546}{8} = 68.25$$

$$\bar{y} = \frac{67 + 68 + 64 + 68 + 72 + 70 + 69 + 70}{8}$$

$$= \frac{548}{8} = 68.5$$

$$x \quad y \quad x = x - \bar{x} \quad y = y - \bar{y} \quad x^2 \quad y^2 \quad xy$$

x	y	$x = x - \bar{x}$	$y = y - \bar{y}$	xy	x^2	y^2
65	67	-3.25	-1.5	4.875	10.56	2.25
66	68	-2.25	-0.5	1.125	5.06	0.25
67	64	-1.25	-4.5	5.625	1.56	20.25
67	68	-1.25	-0.5	0.625	1.56	0.25
68	72	-0.25	3.5	-0.875	0.625	12.25
69	70	0.75	1.5	1.125	0.56	2.25
71	69	2.75	0.5	1.375	7.56	0.25
73	70	4.75	1.5	7.125	22.56	2.25
				$\Sigma 21$	$\Sigma 50.047$	$\Sigma 40$



Karl Pearson's coefficient

$$r = \frac{\sum XY}{\sqrt{\sum X^2 \cdot \sum Y^2}}$$

$$= \frac{21}{\sqrt{(60.047)(140)}} = \frac{21}{\sqrt{2001.88}}$$

$$= \frac{21}{44.74} = 0.4693$$

$$\frac{KP - PJ}{E} \approx 0.47$$

b) Given Regression lines are

$$3x + 12y = 19 \quad \text{--- (1)}$$

$$3y + 9x = 16 \quad \text{--- (2)}$$

i. correlation coefficient

$$r = \sqrt{b_{xy} \times b_{yx}}$$

$$\text{from (1)} \quad b_{xy} = \frac{3x + 12y = 19}{3x} = \frac{19 - 12y}{3}$$

$$x = \frac{19 - 12y}{3}$$

$$x = \frac{-12}{3} \cdot \frac{19}{3} - 4y$$

$$\left(\frac{11}{3}\right) b_{xy} = -4 = (v, x)$$

from (2)

 b_{yx}

$$y = \frac{46 - 9x}{3}$$

$$y = \frac{46}{3} - 3x$$

$$b_{yx} = -3$$

$$r = \sqrt{(-4) \times (-3)} = \sqrt{12} = 3.464$$



ii. Mean values of x and y .

Mean value is the point of intersection of two lines

from ① $3x + 12y = 19$

$$3x = 19 - 12y \quad | \quad y = \frac{19 - 3x}{12}$$

$$x = \frac{19 - 12y}{3}$$

from ② $3y + 9x = 46 \Rightarrow y = \frac{46 - 9x}{3}$

$$3y + 9\left(\frac{19 - 12y}{3}\right) = 46$$

$$9y + 171 - 108y = 138$$

$$-99y = -33$$

$$y = \frac{-33}{-99} = \frac{1}{3}$$

$$\Rightarrow x = \frac{19 - 12\left(\frac{1}{3}\right)}{3}$$

$$= \frac{627 - 132}{99} = \frac{495}{99}$$

$\therefore \text{mean} = \left(5, \frac{1}{3}\right)$.

$$(x, y) = \left(\frac{495}{99}, \frac{1}{3}\right)$$

sub y in eqn ②

$$3\left(\frac{19 - 3x}{12}\right) + 9x = 46$$

from ① $y = \frac{19 - 3x}{12}$ $x = \frac{165}{33} = 5$

$$= \frac{4}{12} = \frac{1}{3}$$

$\therefore x, y = \left(5, \frac{1}{3}\right)$.



9a) Given equation $y = a e^{bx}$

$$y = a e^{bx}$$

apply 'ln' on Both sides

$$\ln y = \ln a + bx$$

$$Y = A + BX \quad \text{--- (1)}$$

where $Y = \ln y$, $A = \ln a$

$$B = b \quad x = x$$

$n = 5$

Normal equations of eqⁿ (1) are

$$\sum Y = nA + B \sum x \quad \text{--- (2)}$$

$$\sum xY = A \sum x + B \sum x^2 \quad \text{--- (3)}$$

x	y	Y = ln y	x	xy	x ²
0.0	0.10	-2.30	0.0	0	0
0.5	0.45	-0.79	0.5	-1.29	0.25
1.0	2.15	0.76	1.0	0.76	1
1.5	9.15	2.20	1.5	3.3	2.25
2.0	40.35	3.69	2.0	7.38	4
2.5	180.75	5.19	2.5	12.97	6.25
$\sum x$	$\sum y$	$\sum Y$	$\sum x$	$\sum xy$	$\sum x^2$
7.5	231.75	8.75	7.5	23.125	13.75

Substitute the above values in (2) & (3) eqⁿs

$$8.75 = 5A + B(7.5) \quad \text{--- (4)}$$

$$23.125 = 7.5A + 13.75B \quad \text{--- (5)}$$

Solving (4) & (5), we get

$$A = -4.25 \quad B = 4$$

$$A = \ln a$$

$$a = e^A = e^{-4.25} = 0.0145, \quad b = 4$$



$$\ln y = 0.014 + 4x$$

$$y = e^{0.014} \cdot e^{4x}$$

$$\therefore a = 0.014, b = 4$$

b) Given: x and y values

$$b_{xy} = \frac{\sum xy}{\sum y^2}, \quad b_{yx} = \frac{\sum xy}{\sum x^2}$$

$$x = x - \bar{x}$$

$$y = y - \bar{y}$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{y} = \frac{\sum y}{n}$$

$$\bar{x} = \frac{65 + 66 + 67 + 67 + 68 + 69 + 70 + 72}{8}$$

$$= \frac{544}{8} = 68$$

$$\bar{y} = \frac{67 + 68 + 65 + 68 + 72 + 72 + 69 + 71}{8}$$

$$= \frac{552}{8} = 69$$

x	y	$x = x - \bar{x}$	$y = y - \bar{y}$	xy	x^2	y^2
65	67	-3	-2	6	9	4
66	68	-2	-1	2	4	1
67	65	-1	-4	4	1	16
67	68	-1	-1	1	1	1
68	72	0	3	0	0	9
69	72	1	3	3	1	9
70	69	2	0	0	4	0
72	71	4	2	8	16	4



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